

Near-Square-Root NP-Hardness for Euclidean CVP Would Collapse the Exponential Hierarchy

somewhereafter

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Abstract

Aggarwal, Bennett, Brakerski, Golovnev, Kumar, Li, Peters, Stephens-Davidowitz, and Vaikuntanathan proved a quantitative public-coin protocol placing rank- D Euclidean GapCVP_γ in coAM with complexity $2^{O(D/\gamma^2)}$ up to polynomial factors. We record a consequence at the near-square-root approximation frontier. If, for any fixed $C > 0$, Euclidean

$$\text{GapCVP}_{\sqrt{D}/\log^C D}$$

is NP-hard under polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level. The proof combines the quantitative lattice protocol with the resource-scaled Boppana–Håstad–Zachos theorem of Pavan, Selman, Sengupta, and Vinodchandran.

More generally, for a reduction from inputs of length n producing rank $D(n)$ and approximation $\gamma(D(n))$, the same conclusion follows whenever $D(n)/\gamma(D(n))^2 = \log^{O(1)} n$. Writing $\gamma(D) = \sqrt{D}/f(D)$ exposes the exact protocol frontier: every fixed polylogarithmic loss $f(D) = \log^{O(1)} D$ is conditionally excluded. For rank-regular reductions, the first dimension-only regime not excluded by this argument is the superpolylogarithmic-loss strip $f(D) \notin \log^{O(1)} D$. The genuinely near-square-root portion of that region additionally has $f(D) = D^{o(1)}$. This is a barrier theorem, not an unconditional lower bound and not a construction in the surviving strip.

1 Introduction

The Euclidean closest vector problem asks for the distance from a target $t \in \mathbb{Q}^D$ to a full-rank lattice $L \subseteq \mathbb{R}^D$. Its promise version GapCVP_γ distinguishes

$$\text{dist}(t, L) \leq r \quad \text{from} \quad \text{dist}(t, L) > \gamma(D)r.$$

The square-root scale is a persistent boundary in the complexity of lattice problems, beginning with the classical coAM barrier of Goldreich and Goldwasser [GG00]. A particularly sharp prospective hardness factor is

$$\gamma_C(D) = \frac{\sqrt{D}}{\log^C D}, \tag{1}$$

where $C > 0$ is fixed. This factor is $D^{1/2-o(1)}$ and is asymptotically larger than D^c for every fixed $c < 1/2$. It should be compared with Mira’s formalized NP-hardness theorem at D^c for every fixed $c < 1/2$ [Mir26].

Our main observation is that (1) is already inside a universal quasipolynomial-time coAM regime. This fact has a stronger consequence than a construction-specific obstruction: polynomial-time NP-hardness at (1) would collapse the exponential hierarchy. No classification of lattice gadgets, codes, Fourier witnesses, or source compilers is needed.

The ingredients are published. Aggarwal et al. [ABB⁺26] prove the time–approximation tradeoff

$$\text{GapCVP}_\gamma \in \text{coAM-TIME} \left(2^{O(D/\gamma^2)} \text{poly}(M) \right), \quad (2)$$

where M is the target-instance encoding length. Pavan et al. [PSSV07] prove a resource-scaled extension of the Boppana–Håstad–Zachos collapse theorem [BsZ87]: quasipolynomial Arthur–Merlin protocols for coNP collapse the exponential hierarchy to its second level. Substituting (1) into (2) gives complexity $2^{O(\log^{2^C} D)} \text{poly}(M)$, which is quasipolynomial under any polynomial-time reduction.

The contribution of this note is threefold.

1. We state and prove the resulting architecture-independent conditional impossibility theorem for near-square-root **GapCVP** hardness.
2. We give the source-sensitive form $D(n)/\gamma(D(n))^2 = \log^{O(1)} n$, avoiding an implicit assumption that target rank and source length are polynomially equivalent.
3. We isolate the first asymptotic strip not covered by the quasipolynomial-protocol argument and separate it from the stronger AMETH barrier already implicit in [ABB⁺26].

The theorem is conditional. It does not show unconditionally that the reduction cannot exist. Rather, a correct reduction would itself establish a major and unexpected collapse of complexity classes.

2 Preliminaries

2.1 Promise hardness

We use deterministic polynomial-time promise-preserving many-one reductions. If A is a language and $\Pi = (\Pi_Y, \Pi_N)$ is a promise problem, a reduction R satisfies

$$x \in A \Rightarrow R(x) \in \Pi_Y, \quad x \notin A \Rightarrow R(x) \in \Pi_N.$$

We call Π NP-hard if such a reduction exists from an NP-complete language.

Let n denote source length, $M(n)$ the bit length of the emitted lattice instance, and $D(n)$ its rank. Polynomial time implies

$$M(n) \leq n^{O(1)}, \quad D(n) \leq M(n). \quad (3)$$

We do not assume a polynomial lower bound on $D(n)$ unless stated.

2.2 Quasipolynomial and exponential resources

Write

$$\text{qpoly}(n) = 2^{(\log n)^{O(1)}}.$$

The class AM_{qpoly} consists of languages with constant-round public-coin protocols whose verifier time, randomness, and total communication are quasipolynomial. Define PH_{qpoly} by allowing a fixed number of alternations with quasipolynomial resources.

The exponential hierarchy is

$$\text{EH} = \bigcup_{k \geq 1} \Sigma_k^{\text{EXP}}.$$

We use the following published collapse theorem.

Theorem 2.1 (Pavan–Selman–Sengupta–Vinodchandran [PSSV07]) *If $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$, then*

$$\text{PH}_{\text{qpoly}} \subseteq \text{AM}_{\text{qpoly}}$$

and, by padding,

$$\text{EH} \subseteq \text{AM}_{\text{EXP}}.$$

The standard containment $\text{AM}_{\text{EXP}} \subseteq \Sigma_2^{\text{EXP}} \cap \Pi_2^{\text{EXP}}$, together with $\Sigma_2^{\text{EXP}} \subseteq \text{EH}$, therefore gives

$$\text{EH} = \text{AM}_{\text{EXP}} \subseteq \Pi_2^{\text{EXP}}.$$

Consequently the exponential hierarchy collapses to its second level.

This strengthens an earlier theorem of Selman and Sengupta that obtained only a third-level collapse. The distinction is important: the later argument extends the Boppana–Håstad–Zachos proof to quasipolynomial resource bounds before applying padding.

3 The quantitative coAM protocol

Aggarwal et al. prove the following tradeoff.

Theorem 3.1 (Aggarwal et al.) *Corollary 3.3 of [ABB⁺26] states that for every $\gamma = \gamma(D) \geq 1$, rank- D Euclidean GapCVP $_\gamma$ has a constant-round coAM protocol of complexity*

$$2^{O(D/\gamma^2)} \text{poly}(M), \tag{4}$$

where M is the input encoding length.

The underlying private-coin protocol performs

$$N_2 = 10D^{3/2}(1 - 1/\gamma^2)^{-(D+1)/2} \tag{5}$$

independent trials. In each trial Arthur chooses a hidden uniform bit and a uniform point in one of two translated Euclidean balls; Merlin must identify the hidden translate. Generic private-to-public-coin and round-reduction transformations incur polynomial overhead, preserving the exponent in (4).

For $\gamma(D) = \sqrt{D}/f(D)$, (4) becomes

$$2^{O(f(D)^2)} \text{poly}(M). \tag{6}$$

Equation (6) is the only lattice-specific input needed below.

4 The hierarchy-collapse barrier

We first state the source-sensitive theorem.

Theorem 4.1 (General reduction barrier) *Suppose there is a deterministic polynomial-time promise-preserving many-one reduction from SAT instances of length n to Euclidean GapCVP $_\gamma$ instances of rank $D(n)$ and encoding length $M(n)$. If, for some fixed a ,*

$$\frac{D(n)}{\gamma(D(n))^2} \leq (\log n)^a \tag{7}$$

for all sufficiently large n , then

$$\text{EH} = \text{AM}_{\text{EXP}} \subseteq \Pi_2^{\text{EXP}}.$$

In particular, the exponential hierarchy collapses to its second level.

Proof. By Theorem 3.1, the target promise problem has a coAM protocol of complexity

$$2^{O(D(n)/\gamma(D(n))^2)} \text{poly}(M(n)).$$

Equations (3) and (7) make this quantity quasipolynomial in n . Composing the verifier with the deterministic reduction gives

$$\text{SAT} \in \text{coAM}_{\text{qpoly}},$$

and complementing gives $\text{TAUT} \in \text{AM}_{\text{qpoly}}$. Since TAUT is coNP -complete under polynomial-time many-one reductions,

$$\text{coNP} \subseteq \text{AM}_{\text{qpoly}}.$$

The conclusion follows from Theorem 2.1. $\square$

Corollary 4.2 (Fixed-polylogarithmic loss) *For every fixed $C > 0$, if*

$$\text{GapCVP}_{\sqrt{D}/\log^C D}$$

is NP-hard under polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level.

Proof. Here

$$\frac{D}{\gamma(D)^2} = \log^{2C} D.$$

Since $D(n) \leq M(n) \leq n^{O(1)}$, this is $\log^{O(1)} n$, so Theorem 4.1 applies. $\square$

Corollary 4.3 (Conditional impossibility) *Assume that the exponential hierarchy does not collapse to its second level. Then, for every fixed $C > 0$, Euclidean $\text{GapCVP}_{\sqrt{D}/\log^C D}$ is not NP-hard under deterministic polynomial-time promise-preserving many-one reductions.*

This conclusion is independent of the internal form of the reduction. It applies equally to reductions built from codes, PCPs, Fourier statistics, group actions, geometric gadgets, or any other polynomial-time compiler.

5 The exact asymptotic frontier of this argument

Write the approximation factor as

$$\gamma(D) = \frac{\sqrt{D}}{f(D)}, \tag{8}$$

where $f(D)$ is the loss from the square-root scale. Theorem 4.1 becomes the following exact criterion.

Proposition 5.1 (Protocol-frontier criterion) *A polynomial-time reduction with loss f implies the second-level exponential-hierarchy collapse whenever*

$$f(D(n))^2 = \log^{O(1)} n. \quad (9)$$

In particular, every dimension-only bound

$$f(D) = \log^{O(1)} D \quad (10)$$

satisfies (9) for every polynomial-time reduction.

The converse is deliberately not claimed. If $f(D)$ is superpolylogarithmic in D , condition (9) may still hold when the reduction emits unusually small rank relative to source length. To state a clean dimension-only boundary, one may impose the natural rank-regularity condition

$$D(n) \geq n^\delta \quad (11)$$

for some fixed $\delta > 0$.

Corollary 5.2 (Rank-regular boundary) *Under (11), the protocol in (4) has quasipolynomial complexity precisely when*

$$f(D) = \log^{O(1)} D$$

along the output ranks, up to the usual eventual-bound interpretation. Thus the first dimension-only region not covered by the EH-noncollapse argument consists of losses not bounded by any fixed polylogarithm. The near-square-root portion of this region is

$$\log^C D = o(f(D)) \text{ for every fixed } C, \quad f(D) = D^{o(1)}, \quad \gamma(D) = \frac{\sqrt{D}}{\log^{\omega(1)} D}. \quad (12)$$

There is no smallest superpolylogarithmic function. Equation (12) describes a boundary class, not one canonical approximation factor. Examples include

$$f(D) = \exp((\log \log D)^{1+\varepsilon}) \quad \text{and} \quad f(D) = \exp(\sqrt{\log D}).$$

Both give $\gamma(D) = D^{1/2-o(1)}$, which is asymptotically larger than D^c for every fixed $c < 1/2$, while the protocol in (6) is superquasipolynomial.

Failure of the present argument in (12) is not evidence that hardness exists there. It says only that the published quantitative coAM protocol no longer places the composed source problem in AM_{qpoly} .

6 The stronger AMETH boundary

Aggarwal et al. define AMETH as the assertion that, for some $\varepsilon > 0$, 3-TAUT has no constant-round public-coin protocol of complexity $2^{\varepsilon n}$. The same composition gives a stronger but less structural conditional barrier. Here, unlike in Sections 2–5, n denotes the number of variables. We restrict the proposed reduction to 3-CNF formulas with $O(n)$ clauses and let $D(n)$ be the largest emitted rank on those inputs. Their encoding length is $O(n)$; the standard sparsification lemma [IPZ01] lifts any $2^{o(n)}$ protocol for this sparse restriction to a $2^{\delta n + o(n)}$ protocol for general 3-TAUT, for every fixed $\delta > 0$.

Proposition 6.1 (AMETH barrier) *Under AMETH, no reduction as in Theorem 4.1 can satisfy*

$$\frac{D(n)}{\gamma(D(n))^2} = o(n). \quad (13)$$

on all 3-CNF inputs with n variables and $O(n)$ clauses. Equivalently, for (8), it cannot satisfy

$$f(D(n))^2 = o(n). \quad (14)$$

Proof. Equations (4) and (13) give a $2^{o(n)}$ constant-round protocol for sparse 3-TAUT. Apply the sparsification lemma and run the resulting sparse protocols in parallel. For every fixed $\delta > 0$, this gives a constant-round protocol of complexity $2^{\delta n + o(n)}$ for general 3-TAUT. Taking δ below the constant in AMETH yields a contradiction. $\square$

If $D(n) \leq n^K$ for fixed K and $f(D) = D^{o(1)}$, then $f(D(n))^2 = n^{o(1)} = o(n)$. Hence AMETH excludes the entire $D^{1/2-o(1)}$ approximation regime for polynomial-output reductions, not only fixed-polylogarithmic losses. The superpolylogarithmic strip (12) is therefore open only relative to the weaker EH-noncollapse assumption; it is already forbidden by AMETH.

For a fixed-power factor $\gamma(D) = D^c$, the protocol exponent is D^{1-2c} . If $D(n) \leq n^K$, AMETH forces the efficiency ceiling

$$K(1 - 2c) \not\leq 1, \quad \text{or heuristically} \quad c \leq \frac{1}{2} - \frac{1}{2K}, \quad (15)$$

unless constants or lower-order behavior keep (13) from holding. Thus hardness for every fixed $c < 1/2$ can coexist with the barrier when approaching $1/2$ requires an unbounded output-degree parameter; a uniform $D^{1/2-o(1)}$ theorem cannot.

7 Why the protocol does not give a deterministic algorithm

The conclusion of Corollary 4.2 is an interactive-proof collapse, not a quasipolynomial-time algorithm for GapCVP. This distinction cannot be removed by naively enumerating Arthur’s random coins.

The private protocol underlying (4) uses N_2 independent hidden bits even before accounting for the random convex-body samples. Therefore its written implementation consumes at least N_2 random bits. At the near-square-root endpoint, substituting $\gamma = \sqrt{D}/f(D)$ into (5) gives

$$\log N_2 = \log 10 + \frac{3}{2} \log D + \frac{1}{2} f(D)^2 + O\left(\frac{f(D)^2 + f(D)^4}{D}\right). \quad (16)$$

Thus $N_2 = 2^{\log^{O(1)} D}$ for polylogarithmic f , and exhaustive enumeration costs

$$2^{\Omega(N_2)},$$

not quasipolynomial time. Corollary 3.3 of [ABB⁺26] invokes generic Goldwasser–Sipser [GS89] and Babai–Moran [BM88] transformations with polynomial overhead; it does not replace the experiment by a pairwise-independent test with a logarithmic seed.

More precisely, let ℓ be the amplified private random-tape length in the Goldwasser–Sipser simulation. Its explicit public-coin implementation uses full random linear maps and at least $2\ell^2$ public random bits at an intermediate stage. Since $\ell \geq N_2$, enumerating those public strings already costs $2^{\Omega(N_2^2)}$. The final Babai–Moran transformation is stated only with polynomial resource

overhead and supplies no alternative polylogarithmic seed bound. Pairwise-universal hashing in the proof therefore does not amount to a short-seed derandomization.

Nor can a standard pseudorandom generator simply be substituted. In a public-coin protocol Merlin’s response depends on the revealed coins, and soundness quantifies over all prover strategies. Fooling a fixed space-bounded computation is not by itself a derandomization of this interactive maximum. Accordingly, this note claims $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$ under the hypothetical hardness reduction, but not $\text{coNP} \subseteq \text{QP}$ and not a deterministic quasipolynomial-time algorithm for GapCVP .

One may obtain the standard nonuniform containment

$$\text{coAM}_{\text{qpoly}} \subseteq \text{coNP}/\text{qpoly}$$

by fixing a quasipolynomial-size advice list of public strings after amplification and a union bound over all inputs of a given length. The advice slash is essential; this is not a uniform seed-enumeration algorithm and is not used in our main theorem.

8 Relation to fixed-exponent hardness

Mira (@Mira__Mira_) proves NP-hardness at D^c for every fixed $c < 1/2$ [Mir26]. This theorem does not diagonalize to (1): each fixed exponent corresponds to polynomial loss

$$f(D) = D^{1/2-c},$$

whereas (1) has only polylogarithmic loss. A family of separate reductions, one for each fixed c , need not yield a single polynomial-time reduction with $c = c(D) \rightarrow 1/2$. Typically the polynomial degree of the output size depends on c and becomes unbounded as $c \rightarrow 1/2$.

Under EH noncollapse, Corollary 4.2 says that this distinction is not merely a defect of one known construction: no polynomial-time Karp reduction can reach the fixed-polylogarithmic-loss endpoint. It does not rule out an improvement into the superpolylogarithmic strip (12).

9 Conclusion

The quantitative coAM protocol for Euclidean GapCVP and the resource-scaled interactive-proof collapse theorem together give a universal conditional boundary:

$$\boxed{\text{NP-hardness at } \sqrt{D}/\log^{O(1)} D \implies \text{the exponential hierarchy collapses to level two.}}$$

This makes any such hardness proof simultaneously a hierarchy-collapse proof. Assuming EH noncollapse, the fixed-polylogarithmic-loss endpoint is closed.

The next unresolved asymptotic target, for rank-regular reductions and under that assumption alone, is hardness with a loss that is simultaneously superpolylogarithmic and subpolynomial, schematically

$$\frac{\sqrt{D}}{\log^{\omega(1)} D}.$$

This strip lies strictly beyond every fixed power D^c , $c < 1/2$, but outside the quasipolynomial- coAM zone. Establishing hardness there would require a new reduction; excluding it would require an assumption or upper bound stronger than EH noncollapse.

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